Chapter 1: Linear Systems

Exercise Set Part 1 of 5

Show your reasoning and row operations. Use exact fractions where possible. Unless stated otherwise, solve by hand; software may be used to check your results.

A. Row, column, and matrix pictures

1. For the system
$$x - 2y = 4, \qquad 3x + y = 5,$$
(a) write $A\mathbf{x} = \mathbf{b}$; (b) write the column-vector equation; (c) solve the system; and (d) sketch the two lines and label their intersection.

2. Let $A = \begin{pmatrix} 1 & 2 & -1 \\ 0 & 3 & 4 \\ 2 & -1 & 1 \end{pmatrix}$ and $\mathbf{v} = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$. Compute $A\mathbf{v}$ in two ways: using row dot products and using a linear combination of columns. Explain why the answers agree.

3. Interpret each system geometrically and determine its solution set:
$$\text{(a) } x + y = 2, \quad 2x + 2y = 4; \qquad \text{(b) } x + y = 2, \quad 2x + 2y = 5.$$
For each case, explain whether the right-hand side is a linear combination of the columns of the coefficient matrix.

4. Consider $x + y + z = 3$, $x - y + z = 1$, and $2x + z = 4$. Solve the system and describe the intersection of the three planes. Verify your answer by forming the corresponding linear combination of columns.

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B. Gaussian elimination and zero pivots

5. Use Gaussian elimination and back substitution to solve

$$\begin{aligned} 2x + y - z &= 2, \\ 4x + 5y - 3z &= 8, \\ -2x + 5y - 2z &= 1. \end{aligned}$$

   Show the augmented matrix, each row operation, and the three pivots.

6. Solve $2y + z = 5$, $x + y = 3$, $2x + 3y + 4z = 16$ by elimination. Start by exchanging the first and second rows. Explain why a zero in the original first pivot position does not imply that the matrix is singular.

7. Apply elimination to $x + 2y - z = 1$, $2x + 4y - 2z = 2$, and $-x - 2y + z = 3$. State whether a solution exists. If the last right-hand side is changed from 3 to -1, give all solutions.

8. For real parameters a and b, classify the system

$$x + 2y = 3, \qquad 2x + ay = b$$

   as having a unique solution, infinitely many solutions, or no solution. Give the solution or a parametrization whenever one exists.

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C. Elementary and permutation matrices

9. For a matrix with three rows, construct the elementary matrix E for $R_3 \leftarrow R_3 - 4R_1$. Find E^{-1} and verify $E^{-1}E = I$. Explain why EA performs this operation on any matrix A with three rows.

10. For the coefficient matrix in Exercise 5, construct E_1, E_2, and E_3 for the three elimination operations, in that order. Compute $E_3E_2E_1A$ and explain why the order matters.

11. Let $P = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$. Compute PB and P^{-1}. Describe the row permutation and verify $P^{-1}P = I$.

12. Give a short proof that $P^{-1} = P^T$ for every permutation matrix P. Then give a permutation matrix of size 3×3 that is not its own inverse, and explain why applying the same permutation twice does not undo it.

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D. LU factorization and pivoting

13. Find $A = LU$ for the coefficient matrix in Exercise 5, with L lower triangular and all diagonal entries equal to 1. Verify the product. Use forward and back substitution to solve the system in Exercise 5 again.

14. Let $B = \begin{pmatrix} 0 & 2 & 1 \\ 1 & 1 & 0 \\ 2 & 3 & 4 \end{pmatrix}$. Find $PB = LU$ by first exchanging rows 1 and 2, then eliminating without further exchanges. Record P, L, and U, and verify the factorization. This prescribed exchange need not follow the partial-pivoting rule.

15. Consider $A\mathbf{x} = \mathbf{b}$ with

$$A = \begin{pmatrix} 0.01 & 1 \\ 1 & 1 \end{pmatrix}, \qquad \mathbf{b} = \begin{pmatrix} 1.01 \\ 2 \end{pmatrix}.$$

Using exact arithmetic, find the first elimination multiplier without pivoting and the first multiplier with partial pivoting. For the pivoted method, give P, L, and U, then solve the system. Explain the numerical motivation for partial pivoting.

16. Reuse the factors from Exercise 13 to solve $A\mathbf{w} = \mathbf{c}$, where $\mathbf{c} = (3, 7, 0)^T$. Show both triangular solves. Explain why reusing the factors is preferable to repeating elimination for every new right-hand side.

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E. Inverses and computational cost

17. Use Gauss–Jordan elimination on $[D \mid I]$ to find the inverse of $D = \begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix}$. Show your row operations, verify $DD^{-1} = I$, and use the inverse to solve $D\mathbf{x} = (1, 0)^T$.

18. Use Gauss–Jordan elimination to find C^{-1} for $C = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix}$. Show the main augmented-matrix stages and verify your result by multiplication.

19. Let A and B be invertible square matrices of the same size. Verify that $B^{-1}A^{-1}$ is the inverse of AB by multiplying in both orders. For $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$, show explicitly that $A^{-1}B^{-1}$ is not the inverse of AB.

20. For a dense $n \times n$ system, assume factorization costs approximately $\frac{2}{3}n^3$ arithmetic operations and a pair of triangular solves costs approximately $2n^2$. (a) Estimate the total cost for k right-hand sides when the factors are reused. (b) Estimate the cost when factorization is repeated for each right-hand side. (c) Evaluate both estimates for $n = 100$ and $k = 10$, and find their ratio. (d) Approximately how does the factorization cost change when n doubles? Why might a tridiagonal system be cheaper?